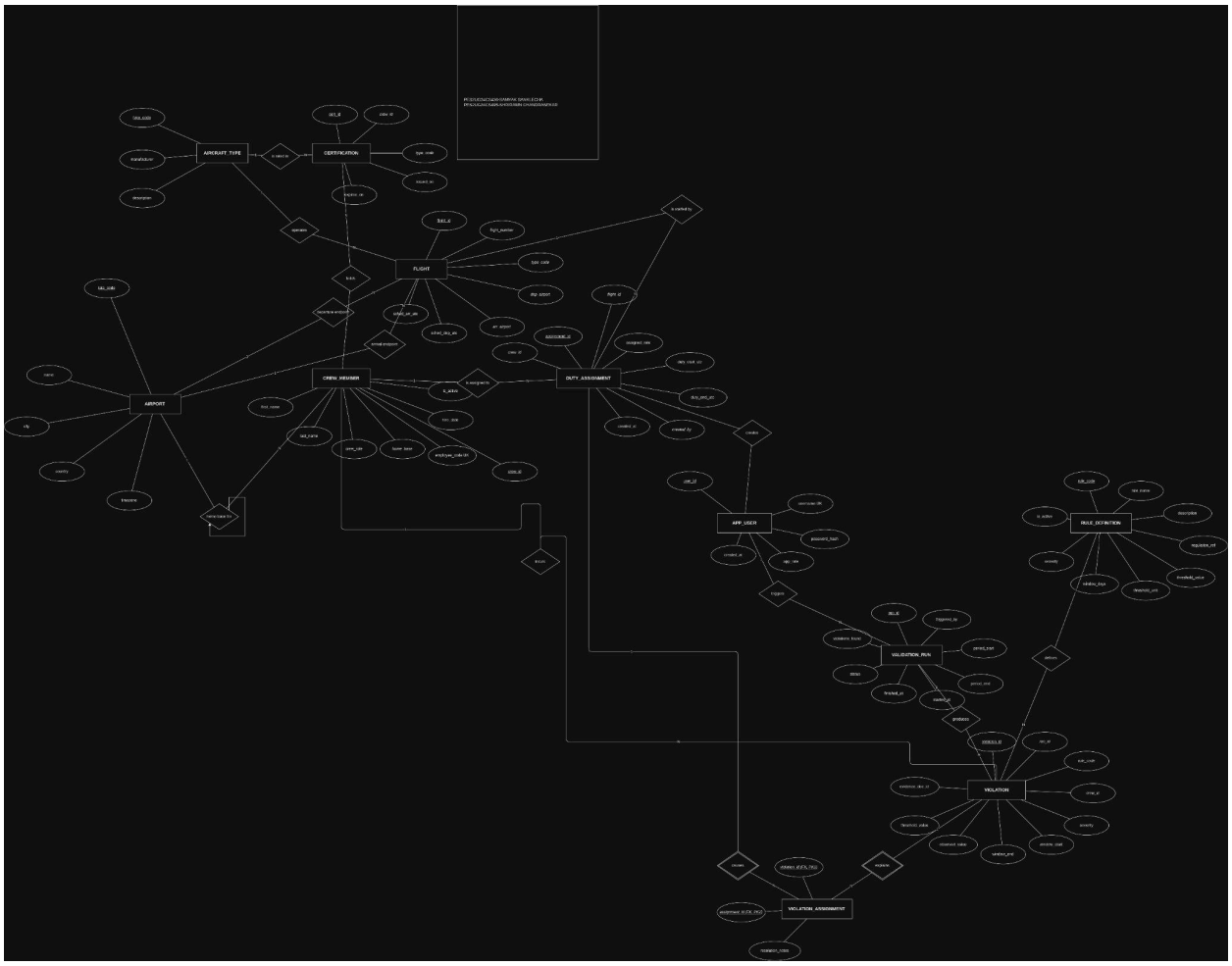
DBMS MINIPROJECT

GROUP-
PES2UG24CS436:SAMYAK SANKELCHA
PES2UG24CS495:SHRIRAM CHANDRASEKAR

PROJECT TITLE:Regulatory Compliance Validation and Violation Explanation for Airline Crew Rosters

PROJECT ABSTRACT:
Even when staffing is at capacity, they can be illegal as aviation authorities have limits on working hours for crews and require rest periods and type ratings. These rules are not detected by a single flight, but are detectable in terms of overlapping time frames such as accumulated working hours during any week. CrewGuard applies a complete crew schedule to five such regulations in temporal SQL on PostgreSQL database and identifies those flights contributing to the above violations. MongoDB has a flexible schema for capturing rule specific evidence.

ER DIAGRAM:



For better clarity we have provided a link below to access the ER diagram:

link

DDL COMMANDS -

```
postgres=# ^C
PES2UG24CS436/495=> CREATE DATABASE crewguard;
CREATE DATABASE
```

```
You are now connected to database "crewguard" as user "samyaksanklecha".
PES2UG24CS436/495=> CREATE EXTENSION IF NOT EXISTS btree_gist;
CREATE EXTENSION
PES2UG24CS436/495=> CREATE TABLE airport (
crewguard(# iata_code       CHAR(3)      PRIMARY KEY,
    name            VARCHAR(120) NOT NULL,
    city            VARCHAR(80)  NOT NULL,
    country         VARCHAR(80)  NOT NULL,
    timezone        VARCHAR(64)  NOT NULL,
    CONSTRAINT chk_airport_iata_format CHECK (iata_code ~ '^[A-Z]{3}$')
);
CREATE TABLE
```

```
CREATE TABLE
PES2UG24CS436/495=> CREATE TABLE aircraft_type (
    type_code       VARCHAR(10)  PRIMARY KEY,
    manufacturer    VARCHAR(60)  NOT NULL,
    description     VARCHAR(160),
    CONSTRAINT chk_aircraft_type_upper CHECK (type_code = UPPER(type_code))
);
CREATE TABLE
```

```
PES2UG24CS436/495=> CREATE TABLE app_user (
    user_id         SERIAL       PRIMARY KEY,
    username        VARCHAR(50)  NOT NULL UNIQUE,
    password_hash   VARCHAR(255) NOT NULL,
    app_role        VARCHAR(20)  NOT NULL,
    created_at      TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    CONSTRAINT chk_app_role CHECK (app_role IN ('SCHEDULER', 'AUDITOR'))
);
CREATE TABLE
```

```
PES2UG24CS436/495=> CREATE TABLE crew_member (
    crew_id         SERIAL       PRIMARY KEY,
    employee_code   VARCHAR(20)  NOT NULL UNIQUE,
    first_name      VARCHAR(60)  NOT NULL,
    last_name       VARCHAR(60)  NOT NULL,
    crew_role       VARCHAR(20)  NOT NULL,
    home_base       CHAR(3)      NOT NULL,
    hire_date       DATE         NOT NULL,
    is_active       BOOLEAN      NOT NULL DEFAULT TRUE,
    CONSTRAINT fk_crew_home_base FOREIGN KEY (home_base) REFERENCES airport(iata_code)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT chk_crew_role CHECK (crew_role IN ('CAPTAIN', 'FIRST_OFFICER', 'CABIN_CREW')),
    CONSTRAINT chk_crew_hire_date CHECK (hire_date <= CURRENT_DATE)
);
CREATE TABLE
```

```
PES2UG24CS436/495=> CREATE TABLE certification (
    cert_id          SERIAL       PRIMARY KEY,
    crew_id          INTEGER      NOT NULL,
    type_code        VARCHAR(10)  NOT NULL,
    issued_on        DATE         NOT NULL,
    expires_on       DATE         NOT NULL,
    CONSTRAINT fk_cert_crew FOREIGN KEY (crew_id) REFERENCES crew_member(crew_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_cert_type FOREIGN KEY (type_code) REFERENCES aircraft_type(type_code)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT uk_cert_crew_type_issue UNIQUE (crew_id, type_code, issued_on),
    CONSTRAINT chk_cert_validity_order CHECK (expires_on > issued_on)
);
CREATE TABLE
```

```
PES2UG24CS436/495=> CREATE TABLE flight (
    flight_id       SERIAL       PRIMARY KEY,
    flight_number   VARCHAR(10)  NOT NULL,
    type_code       VARCHAR(10)  NOT NULL,
    dep_airport     CHAR(3)      NOT NULL,
    arr_airport     CHAR(3)      NOT NULL,
    sched_dep_utc   TIMESTAMPTZ  NOT NULL,
    sched_arr_utc   TIMESTAMPTZ  NOT NULL,
    CONSTRAINT fk_flight_type FOREIGN KEY (type_code) REFERENCES aircraft_type(type_code)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_flight_dep FOREIGN KEY (dep_airport) REFERENCES airport(iata_code)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_flight_arr FOREIGN KEY (arr_airport) REFERENCES airport(iata_code)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT uk_flight_number_dep UNIQUE (flight_number, sched_dep_utc),
    CONSTRAINT chk_flight_time_order CHECK (sched_arr_utc > sched_dep_utc),
    CONSTRAINT chk_flight_distinct_endpoints CHECK (dep_airport <> arr_airport)
);
CREATE TABLE
```

```
PES2UG24CS436/495=> CREATE TABLE duty_assignment (
    assignment_id   SERIAL       PRIMARY KEY,
    crew_id         INTEGER      NOT NULL,
    flight_id       INTEGER      NOT NULL,
    assigned_role   VARCHAR(20)  NOT NULL,
    duty_start_utc  TIMESTAMPTZ  NOT NULL,
    duty_end_utc    TIMESTAMPTZ  NOT NULL,
    created_by      INTEGER,
    created_at      TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    CONSTRAINT fk_duty_crew FOREIGN KEY (crew_id) REFERENCES crew_member(crew_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_duty_flight FOREIGN KEY (flight_id) REFERENCES flight(flight_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_duty_created_by FOREIGN KEY (created_by) REFERENCES app_user(user_id)
        ON UPDATE CASCADE ON DELETE SET NULL,
    CONSTRAINT uk_duty_crew_flight UNIQUE (crew_id, flight_id),
    CONSTRAINT chk_duty_time_order CHECK (duty_end_utc > duty_start_utc),
    CONSTRAINT chk_duty_assigned_role CHECK (assigned_role IN ('CAPTAIN', 'FIRST_OFFICER', 'CABIN_CREW'))
);
CREATE TABLE
```

```
CREATE TABLE
PES2UG24CS436/495=> ALTER TABLE duty_assignment
    ADD CONSTRAINT excl_duty_no_overlap
    EXCLUDE USING GIST (
        crew_id WITH =,
        tstzrange(duty_start_utc, duty_end_utc) WITH &&
    );
ALTER TABLE
PES2UG24CS436/495=> CREATE TABLE rule_definition (
```

```
ALTER TABLE
PES2UG24CS436/495=> CREATE TABLE rule_definition (
    rule_code       CHAR(3)      PRIMARY KEY,
    rule_name       VARCHAR(120) NOT NULL,
    description     TEXT         NOT NULL,
    regulation_ref  VARCHAR(120),
    threshold_value NUMERIC(8,2),
    threshold_unit  VARCHAR(20),
    window_days     INTEGER,
    severity        VARCHAR(10)  NOT NULL,
    is_active       BOOLEAN      NOT NULL DEFAULT TRUE,
    CONSTRAINT chk_rule_severity CHECK (severity IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')),
    CONSTRAINT chk_rule_window_days CHECK (window_days IS NULL OR window_days > 0)
);
CREATE TABLE
```

```
CREATE TABLE
PES2UG24CS436/495=> CREATE TABLE validation_run (
    run_id          SERIAL       PRIMARY KEY,
    triggered_by    INTEGER,
    period_start    DATE         NOT NULL,
    period_end      DATE         NOT NULL,
    started_at      TIMESTAMPTZ  NOT NULL DEFAULT NOW(),
    finished_at     TIMESTAMPTZ,
    status          VARCHAR(15)  NOT NULL DEFAULT 'RUNNING',
    violations_found INTEGER     NOT NULL DEFAULT 0,
    CONSTRAINT fk_run_user FOREIGN KEY (triggered_by) REFERENCES app_user(user_id)
        ON UPDATE CASCADE ON DELETE SET NULL,
    CONSTRAINT chk_run_status CHECK (status IN ('RUNNING', 'COMPLETED', 'FAILED', 'STALE')),
    CONSTRAINT chk_run_period_order CHECK (period_end >= period_start),
    CONSTRAINT chk_run_violation_count CHECK (violations_found >= 0)
);
CREATE TABLE
```

```
PES2UG24CS436/495=> CREATE TABLE violation (
    violation_id    SERIAL       PRIMARY KEY,
    run_id          INTEGER      NOT NULL,
    rule_code       CHAR(3)      NOT NULL,
    crew_id         INTEGER      NOT NULL,
    severity        VARCHAR(10)  NOT NULL,
    window_start    TIMESTAMPTZ,
    window_end      TIMESTAMPTZ,
    observed_value  NUMERIC(8,2),
    threshold_value NUMERIC(8,2),
    evidence_doc_id VARCHAR(24),
    CONSTRAINT fk_violation_run FOREIGN KEY (run_id) REFERENCES validation_run(run_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_violation_rule FOREIGN KEY (rule_code) REFERENCES rule_definition(rule_code)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_violation_crew FOREIGN KEY (crew_id) REFERENCES crew_member(crew_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT chk_violation_severity CHECK (severity IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')),
    CONSTRAINT chk_violation_window_order CHECK (window_end IS NULL OR window_start IS NULL OR window_end >= window_start)
);
CREATE TABLE
```

```
CREATE TABLE
PES2UG24CS436/495=> CREATE TABLE violation_assignment (
    violation_id        INTEGER      NOT NULL,
    assignment_id       INTEGER      NOT NULL,
    contribution_value  NUMERIC(8,2),
    CONSTRAINT pk_violation_assignment PRIMARY KEY (violation_id, assignment_id),
    CONSTRAINT fk_va_violation FOREIGN KEY (violation_id) REFERENCES violation(violation_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_va_assignment FOREIGN KEY (assignment_id) REFERENCES duty_assignment(assignment_id)
        ON UPDATE CASCADE ON DELETE CASCADE
);
CREATE TABLE
```

```
             ^
PES2UG24CS436/495=> \dt
                    List of relations
  Schema |          Name          | Type  |      Owner
 --------+------------------------+-------+------------------
  public | aircraft_type          | table | samyaksanklecha
  public | airport                | table | samyaksanklecha
  public | app_user               | table | samyaksanklecha
  public | certification          | table | samyaksanklecha
  public | crew_member            | table | samyaksanklecha
  public | duty_assignment        | table | samyaksanklecha
  public | flight                 | table | samyaksanklecha
  public | rule_definition        | table | samyaksanklecha
  public | validation_run         | table | samyaksanklecha
  public | violation              | table | samyaksanklecha
  public | violation_assignment   | table | samyaksanklecha
 (11 rows)
```

RELATIONAL DIAGRAM:-

PES2UG24CS436-SAMYAK SANKLECHA
PES2UG24CS495-SHRIRAM CHANDRASEKAR

